Program - 7

Visualization of datasets using Matplotlib and Seaborn

Date : 28.09.2026

Aim

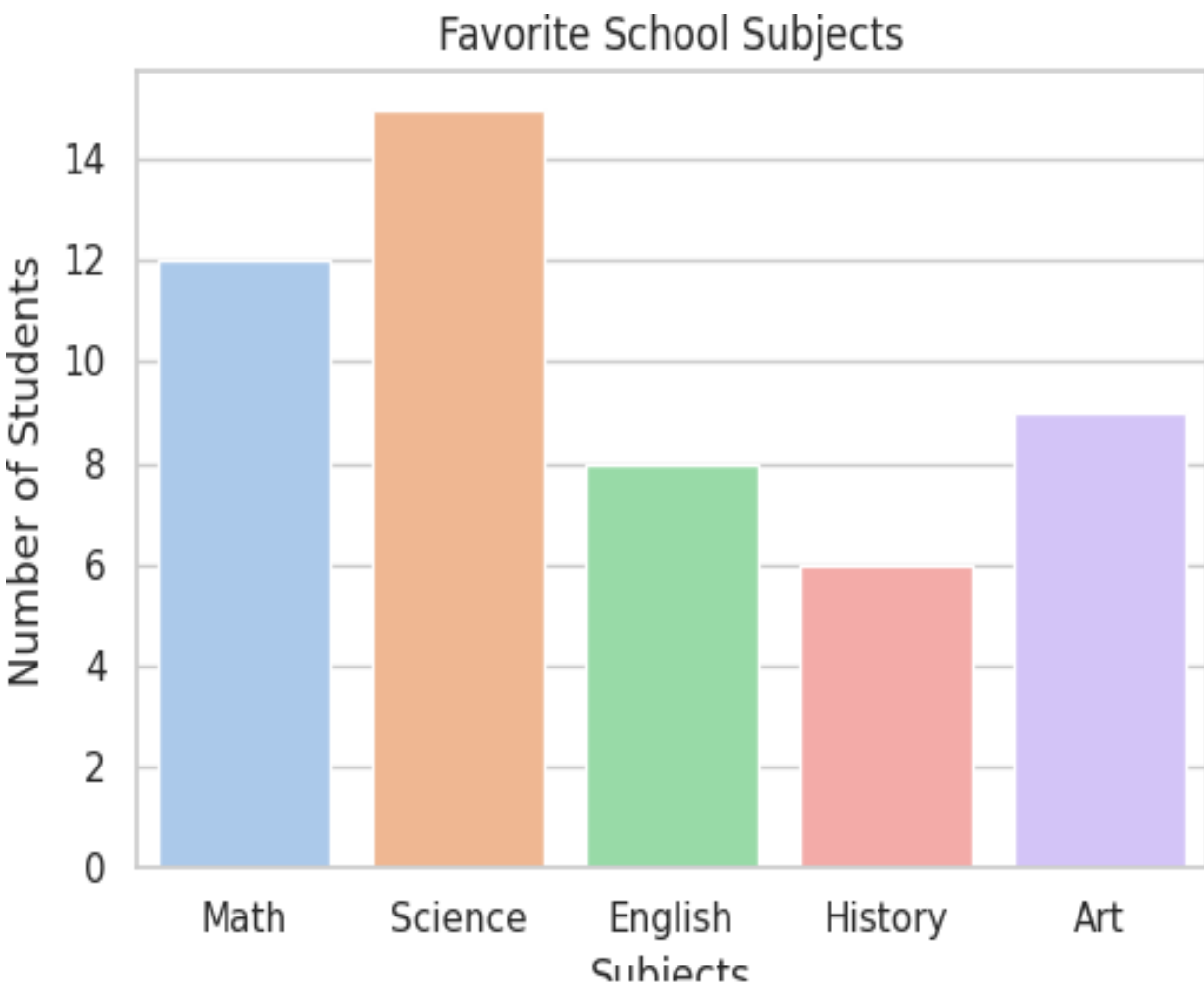
Create a Python program to visualize datasets using Matplotlib and Seaborn by plotting histograms, scatter plots, line charts, and bar graphs.

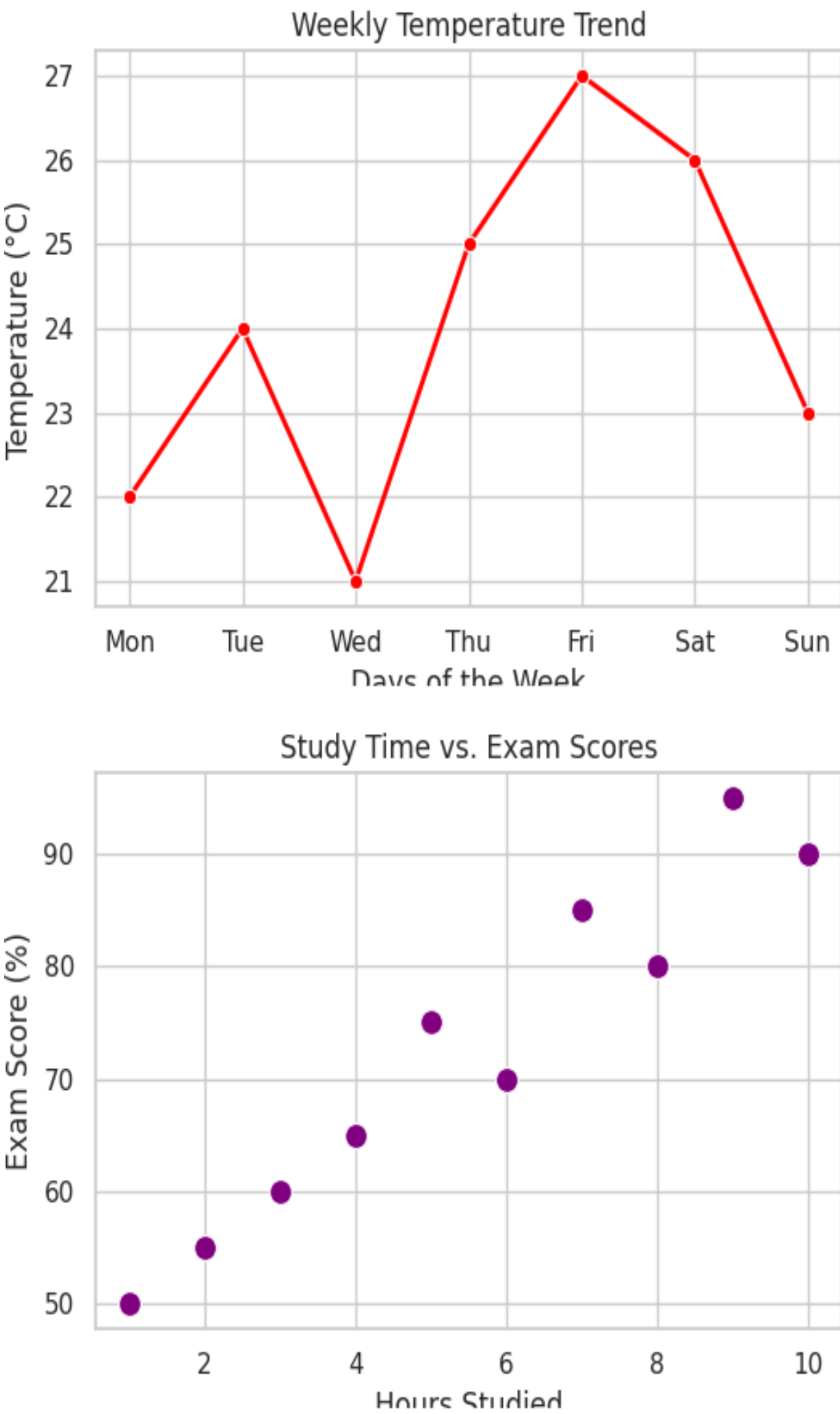
Algorithm

1. Start
2. Import matplotlib.pyplot and seaborn.
3. Set the Seaborn plotting style to whitegrid.
4. Create a Bar Chart:
   - Store the subject names in a list.
   - Store the number of students who prefer each subject.
   - Set the figure size.
   - Create a bar plot using sns.barplot().
   - Add a title, X-axis label, and Y-axis label.
   - Display the bar chart.
5. Create a Line Chart:
   - Store the days of the week in a list.
   - Store the corresponding temperatures.
   - Set the figure size.
   - Create a line plot using sns.lineplot().
   - Add markers to the data points.
   - Add a title and axis labels.
   - Display the line chart.
6. Create a Scatter Plot:
   - Store the number of hours studied.
   - Store the corresponding exam scores.
   - Set the figure size.
   - Create a scatter plot using sns.scatterplot().
   - Add a title and axis labels.
   - Display the scatter plot.
7. Create a Histogram:
   - Store the heights of 20 students.
   - Set the figure size.
   - Create a histogram using sns.histplot() with 5 bins.
   - Add a title and axis labels.
   - Display the histogram.
8. Stop.

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```
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
subjects = ["Math", "Science", "English", "History", "Art"]
students = [12, 15, 8, 6, 9]
plt.figure(figsize=(6, 4))    # Set window size
sns.barplot(x=subjects, y=students, palette="pastel")
plt.title("Favorite School Subjects")
plt.xlabel("Subjects")
plt.ylabel("Number of Students")
plt.show()    # Shows the bar graph
days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
temperature = [22, 24, 21, 25, 27, 26, 23]
plt.figure(figsize=(6, 4))
sns.lineplot(x=days, y=temperature, marker="o", color="red", linewidth=2)
plt.title("Weekly Temperature Trend")
plt.xlabel("Days of the Week")
plt.ylabel("Temperature (°C)")
plt.show()    # Shows the line chart
hours_studied = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
exam_scores = [50, 55, 60, 65, 75, 70, 85, 80, 95, 90]
plt.figure(figsize=(6, 4))
sns.scatterplot(x=hours_studied, y=exam_scores, color="purple", s=100)
plt.title("Study Time vs. Exam Scores")
plt.xlabel("Hours Studied")
plt.ylabel("Exam Score (%)")
plt.show()    # Shows the scatter plot
heights = [140, 145, 142, 150, 155, 152, 153, 160, 165, 147,
                151, 158, 144, 156, 162, 149, 154, 157, 141, 163]
plt.figure(figsize=(6, 4))
sns.histplot(x=heights, bins=5, color="green", kde=False)
plt.title("Distribution of Student Heights")
plt.xlabel("Height Range (cm)")
plt.ylabel("Count")
plt.show()
```


Favorite School Subjects
Number of Students
0
2
4
6
8
10
12
14
Math
Science
English
History
Art
Subjects


Weekly Temperature Trend
Temperature (°C)
21
22
23
24
25
26
27
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Days of the Week

Study Time vs. Exam Scores
Exam Score (%)
50
60
70
80
90
2
4
6
8
10
Hours Studied

Distribution of Student Heights
Count
0
1
2
3
4
5
140
145
150
155
160
165
Height Range (cm)